

Toby Ueno

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RESEARCH INTERESTS

Programming languages, effect handlers, session types, type and effect systems, operational semantics, reactive programming, concurrency

EDUCATION

2025 – present	<i>The University of Edinburgh</i> Ph.D. in Informatics, advised by Dr. Sam Lindley Proposal: Reactive Programming with Effect Handlers
2021 – 2025	<i>Boston University</i> B.A./M.S. in Computer Science B.A. in Linguistics & Philosophy (<i>summa cum laude</i>)

PUBLICATIONS

2026 (ESOP)	T. Ueno and A. Das. Practical Refinement Session Type Inference. <i>In Proceedings of the 35th European Symposium on Programming.</i> • We propose a type inference algorithm for a session type system with arithmetic refinements • We develop a theory of subtyping for session types, including an algorithm which we prove sound with respect to a type simulation-based definition • We implement our algorithm in the Rast language via integration with Z3 • We add key optimizations to make inference feasible, and demonstrate their efficacy via evaluation on challenging examples
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PRESENTATIONS

2025	Practical Refinement Session Type Inference <i>Scottish Programming Languages Seminar</i>
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TEACHING & SERVICE

2025	Coursework Marker <i>INF2-IADS: Introduction to Algorithms & Data Structures</i> The University of Edinburgh
2022 – 2025	Teaching Assistant <i>CS112: Introduction to Computer Science II</i> Boston University
2022 – 2025	President <i>Upsilon Pi Epsilon (Computing & Information Sciences Honor Society)</i> Boston University
2025 – 2025	Course Assistant <i>CS599: Programming Language Foundations for Concurrency</i> Boston University

PROFESSIONAL EXPERIENCE

2024 – 2024	<i>Software Engineering Intern</i> Roblox
2021 – 2025	<i>Full-Stack Software Engineering Intern</i> Fidelity Investments

Last updated December 2025.